

EXPERIMENT 8: DEPTH FIRST SEARCH (DFS)

AIM

To develop a Python program that implements the Depth-First Search (DFS) algorithm for traversing a graph and visiting all reachable nodes.

OBJECTIVE

- To understand the concept of the Depth-First Search (DFS) algorithm.
- To traverse a graph using a depth-first approach.
- To visit each node exactly once.
- To display the DFS traversal order.

ALGORITHM

Step 1: Start.

Step 2: Create a graph using an adjacency list.

Step 3: Initialize an empty set to keep track of visited nodes.

Step 4: Select the starting node.

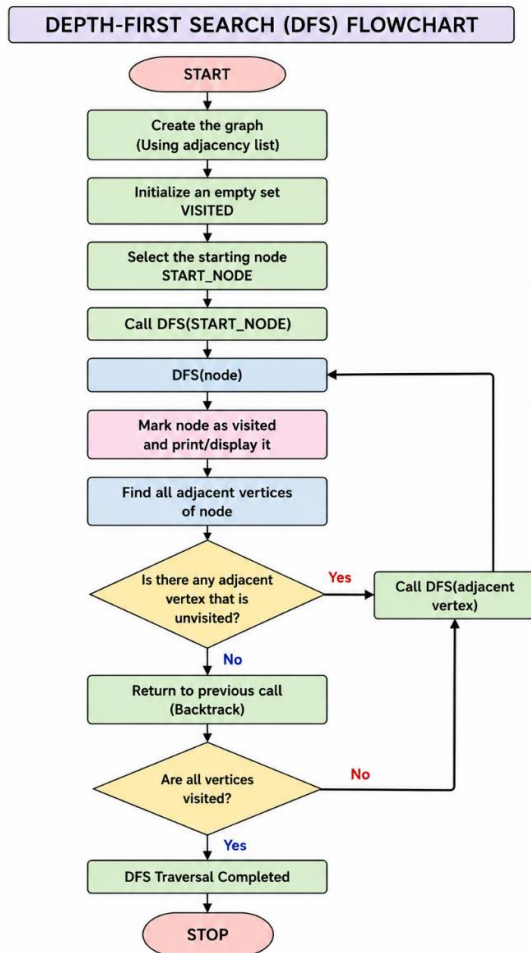
Step 5: Mark the current node as visited and display it.

Step 6: Visit each unvisited adjacent node recursively.

Step 7: Repeat until all reachable nodes are visited.

Step 8: Stop.

FLOW CHART



SOURCE CODE

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}  
visited = set()  
def dfs(node):  
    if node not in visited:  
        print(node, end=" ")  
        visited.add(node)  
  
        for neighbor in graph[node]:  
            dfs(neighbor)  
print("DFS Traversal:")  
dfs('A')
```

OUTPUT

```
DFS Traversal:  
A B D E F C  
|
```

RESULT

The Python program successfully performs Depth-First Search (DFS) on the given graph and displays the traversal order by visiting each node exactly once using the depth-first approach.